

Performance Estimation of Aircraft Computer Vision Models using Optimal Transport-based Distributional Mismatch

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Abstract—Ensuring the reliable deployment of deep learning models in real-world environments remains a fundamental challenge, particularly in safety-critical applications such as autonomous unmanned aerial system (UAS) landings. This paper investigates the relationship between model performance and distributional mismatch between training data and operational design domain (ODD) samples. We propose a framework that quantifies this mismatch using Wasserstein distances derived from optimal transport theory, applied to both raw image data and semantic embeddings from a pretrained CLIP (ViT-B/32) model. To address the scarcity of real ODD data collected by flight tests, we additionally leverage diffusion models to generate additional ODD scenarios, and evaluate their quality using Frchet Inception Distance (FID) score. Experimental results demonstrate a strong correlation between the magnitude of distributional mismatch and model performance degradation. Moreover, our results demonstrate that models trained on datasets with lower distributional mismatch to the ODD domain, such as VisDrone, compared to COCO, achieve superior detection performance. These findings highlight the effectiveness of our framework as a practical and generalizable tool for estimating model reliability in complex real-world deployment scenarios, when trained on different training datasets.

Index Terms—Distributional Mismatch, Optimal Transport, Operational Design Domain, Wasserstein Distance, Diffusion Model.

I. INTRODUCTION

Accurate perception and decision-making are critical for autonomous systems operating in complex, dynamic environments. In particular, safe landing of unmanned aerial systems (UAS) requires robust visual detection of landing zones and obstacles under varying operational conditions [1]–[4]. Despite the remarkable success of deep learning models, especially convolutional neural networks (CNNs), in controlled environments [5], a major challenge remains: models often experience significant performance degradation when deployed on data distributions that differ from their training datasets [6], [7].

Recent research in distributional robustness and domain adaptation has aimed to address the gap between training and deployment scenarios [11]–[14]. However, most existing approaches focus either on confidence estimation or anomaly detection, rather than directly estimating model performance

based on distributional differences. Furthermore, realistic deployment distributions are often unknown or sparsely sampled, making traditional evaluation approaches insufficient. This issue is especially pronounced in aerial drones, where the scarcity of annotated drone-collected data makes it difficult to capture rare but critical scenarios such as occluded landing pads or unexpected dynamic obstacles which, though infrequent, can pose serious risks if unaccounted for during training.

To address these challenges, we propose a structured framework to quantify distributional mismatch between the training dataset and operational design domain (ODD) samples to analyze model performance degradation. Using tools from optimal transport theory [15], specifically Wasserstein distances [16], we rigorously measure the distributional mismatch between datasets. We further leverage generative AI models, in particular Denoising Diffusion Probabilistic Models (DDPMs) [17], to synthesize ODD-like data when real samples are scarce. DDPMs are capable of generating high-resolution images that not only match the statistical structure of the deployment environment but also simulate rare and safety-critical cases that are difficult to capture in real data. This positions the diffusion model as a powerful simulator for stress-testing perception models across edge-case scenarios.

Our experimental results demonstrate a strong correlation between the magnitude of distributional mismatch and model performance, providing actionable insights for model validation and deployment planning. The remainder of this paper is organized as follows: Section II reviews relevant concepts in optimal transport, Wasserstein distance, diffusion models, and U-Net architectures. Section III describes our methodology for measuring distributional mismatch and synthetic data generation. Section IV presents experimental setups and results on real and generated ODD datasets. Section V discusses implications and limitations, and Section VI concludes the paper with future research directions.

II. BACKGROUND

In this section, we review the key concepts foundational to our approach: optimal transport theory and Wasserstein

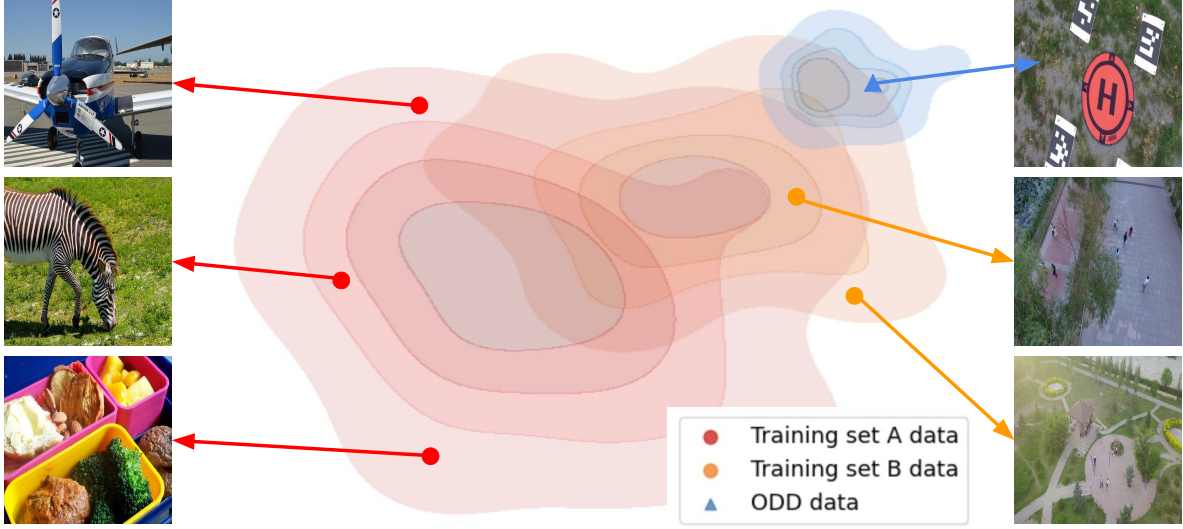


Fig. 1. Conceptual visualization of data distributions in latent space. Each point represents an image sample projected using a CLIP (ViT-B/32) encoder [8]. The red cluster corresponds to the training set A (COCO [9]), the orange cluster to the training set B (VisDrone [10]), and the blue cluster to the operational design domain (ODD) samples. The spatial proximity between VisDrone (orange) and ODD (blue) indicates a lower distributional mismatch compared to COCO (red), which has improved performance of models trained on VisDrone when evaluated on ODD data. This supports our hypothesis that smaller Wasserstein distances correlate with better generalization in deployment scenarios.

distance for quantifying distributional mismatch, and diffusion models with U-Net architectures for synthetic data generation.

A. Optimal Transport Theory and Wasserstein Distance

Optimal Transport (OT) theory provides a rigorous mathematical framework for comparing probability distributions by minimizing the cost of transporting mass from one distribution to another. Initially proposed by Monge [18] and later generalized by Kantorovich [19], OT formulates the comparison between two probability measures under a given cost function.

Given two probability measures μ and ν defined on a metric space $\mathcal{X}$ and a measurable cost function $c : \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$, the optimal transport problem is formulated as:

$$\inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} c(x, y) d\gamma(x, y),$$

where $\Gamma(\mu, \nu)$ denotes the set of all couplings (joint distributions) γ whose marginals are μ and ν respectively, as is shown in Figure 2.

A particularly important case is when the cost function is $c(x, y) = \|x - y\|^p$, leading to the p -Wasserstein distance:

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} \|x - y\|^p d\gamma(x, y) \right)^{1/p}.$$

The Wasserstein distances W_p are all *true metrics* on the space of probability measures with finite p -th moments, satisfying the standard metric axioms: non-negativity, identity of indiscernibles, symmetry, and the triangle inequality [15]. Furthermore, W_p distances can be defined for general types of distributions, including continuous, discrete, and semi-discrete

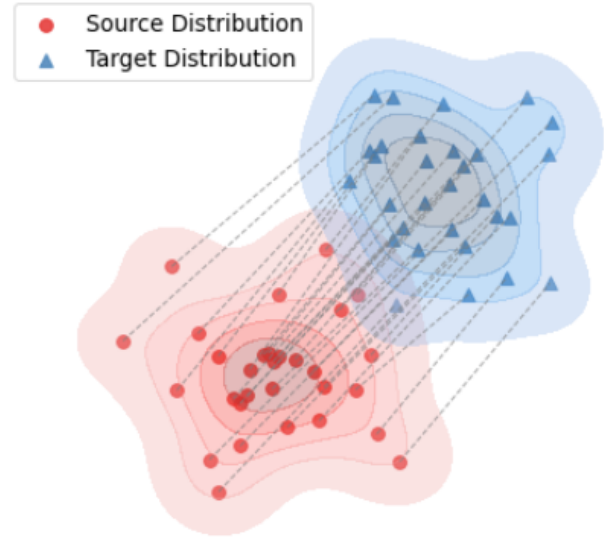


Fig. 2. Illustration of optimal transport between a source and a target distribution. The source distribution (in red) and the target distribution (in blue) are visualized. Dashed gray lines represent the optimal couplings that minimize the transport cost between individual samples from the source to the target distribution.

cases, making them broadly applicable across a wide range of problems in both theory and practice.

In practice, solving OT problems exactly can be computationally intensive. Classical approaches rely on linear programming (LP) formulations [19]. Entropic regularization techniques such as the Sinkhorn algorithm have been introduced [20], enabling faster and more memory-efficient approx-

imations with significantly reduced computational complexity. Additionally, the structure of Sinkhorn iterations lends itself naturally to GPU acceleration, with libraries such as POT [21], GeomLoss [22], and OTT-JAX [23] providing scalable implementations for high-dimensional data.

The choice of p order of the Wasserstein distance directly impacts the behavior and sensitivity of the Wasserstein distance. The 1-Wasserstein distance (W_1) is robust to small perturbations and preserves mass movement over long distances, making it suitable for applications emphasizing spatial shifts. In contrast, the 2-Wasserstein distance (W_2) penalizes larger displacements more heavily and emphasizes variance-like effects between distributions. At the extreme, the Wasserstein-infinity distance (W_∞) captures the maximum displacement needed to map one distribution onto another, providing worst-case guarantees but often at the expense of sensitivity to finer distributional differences. In our experiments, we primarily use W_1 and W_2 due to its interpretability and stability when analyzing operational design domain shifts.

B. U-Net and Diffusion Models

The U-Net architecture [24] is a widely adopted encoder-decoder convolutional network originally designed for biomedical image segmentation. Its symmetric structure with skip connections allows effective multi-scale feature extraction while preserving fine-grained spatial details. Due to these properties, U-Net and its variants have been extensively applied to a wide range of vision tasks, including image denoising [25], super-resolution [26], optical flow estimation [27], and medical image synthesis [28]. These successes demonstrate the U-Net’s versatility in tasks requiring precise spatial reconstruction, making it a natural choice for generative modeling frameworks.

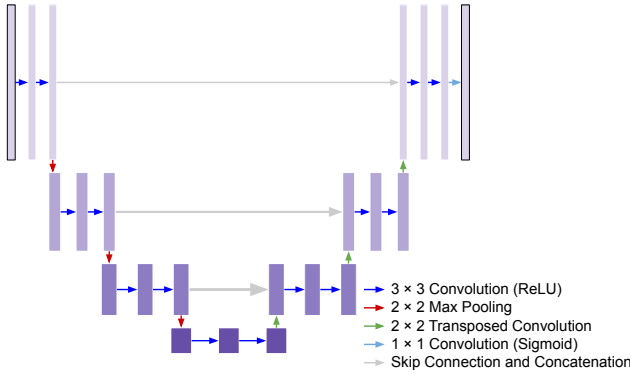


Fig. 3. U-Net architecture is used as the denoising backbone in the diffusion model. The network consists of an encoder-decoder structure with skip connections that preserve spatial resolution and semantic detail. This structure allows the model to progressively remove noise from latent inputs while maintaining fine-grained object information, making it particularly effective for generating high-resolution operational design domain (ODD) scenarios.

Diffusion models [17], [29] leverage the representational power of U-Net backbones to synthesize high-quality data through two stochastic processes: a forward diffusion process

that gradually corrupts data with noise, and a learned reverse process that reconstructs the data. The forward process is typically defined as a Markov chain:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}),$$

where $\beta_t \in (0, 1)$ controls the variance schedule. The generative model then learns to approximate the reverse transitions $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ using a neural network, typically parameterized by a U-Net.

Training is typically simplified by predicting the added noise ϵ , resulting in the following loss function:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2].$$

Beyond the original denoising diffusion probabilistic models (DDPMs) [17], several extensions have been proposed to improve sampling speed and flexibility. For instance, DDIM [30] introduces a deterministic non-Markovian sampling process, significantly accelerating generation. Latent Diffusion Models (LDM) [31] operate in a learned lower-dimensional latent space to improve efficiency without sacrificing sample quality. Improved DDPMs [32] refine noise schedules and network architectures for better performance. These advancements collectively make diffusion models a leading approach in generative modeling.

In this work, we utilize diffusion models trained with U-Net backbones to generate operational design domain (ODD) samples. These generated datasets enable systematic evaluation of model performance under distributional shifts encountered in autonomous landing scenarios.

III. METHODOLOGY

We assume all datasets (both the training set and the ODD set) are empirical samples drawn from their underlying scenario distributions. Specifically, the training dataset defines a *source* distribution, denoted as μ , while the operational design domain (ODD) dataset defines a *target* distribution, denoted as ν . Let M_μ represent a model trained on data from distribution μ , and let $P(M_\mu(\nu))$ denote its performance when evaluated on samples drawn from ν . We characterize the distributional mismatch between the training and ODD datasets as a proxy for the underlying domain mismatch affecting model performance. To measure this mismatch, we compute both 1-Wasserstein (W_1) and 2-Wasserstein (W_2) distance between two distributions. Our hypothesis is that given two different source distributions, μ_1 and μ_2 , if $W_p(\mu_1, \nu) \geq W_p(\mu_2, \nu)$, then the corresponding model performance satisfies $P(M_{\mu_1}(\nu)) \leq P(M_{\mu_2}(\nu))$; That is, larger distributional mismatch correlates with lower model performance in the target domain.

A. Optimal Transport for Performance Comparison

In this subsection, we present a framework for evaluating model performance under distributional mismatch using Optimal Transport. First, we describe how training and operational datasets are represented as empirical distributions, either in

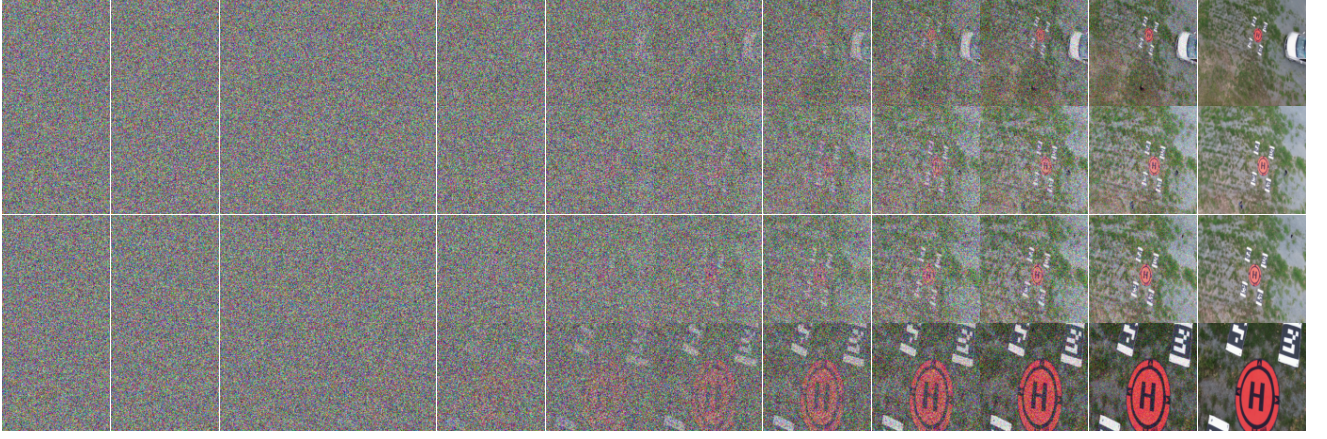


Fig. 4. Visualization of the denoising process in a denoising diffusion probabilistic model (DDPM). Each row represents an independent denoising trajectory, starting from pure Gaussian noise (left) and progressing toward a high-resolution, realistic image (right). The model iteratively refines the sample over a sequence of discrete time steps, gradually removing noise while preserving semantic structure. The final outputs reveal detailed landing pad scenes, including critical features such as the “H” marker and surrounding terrain, illustrating the model’s ability to simulate operational design domain (ODD) scenarios with high fidelity. This demonstrates the utility of DDPMs in generating realistic and rare deployment samples for robustness evaluation.

raw pixel format or using semantically meaningful CLIP embeddings. Then, we detail the computation of Wasserstein distance to quantify the distributional mismatch between these datasets. Finally, we explain how detection performance is assessed using standard mAP metrics.

1) *Data Representation*: We represent each dataset as an empirical distribution. Let the training dataset be denoted as $\mathcal{D}_{\text{train}} = \{x_i\}_{i=1}^N$, and the operational design domain (ODD) dataset as $\mathcal{D}_{\text{odd}} = \{y_j\}_{j=1}^M$, where N and M are the number of the training dataset and ODD dataset, respectively. Each sample x_i and y_j can be represented in one of two formats:

- **Raw format**: Each sample is flattened into a high-dimensional vector in $\mathbb{R}^{d_1}$, where d_1 corresponds to the resolution and channel count of the input images.
- **CLIP-embedding format**: Each sample is passed through a pretrained CLIP encoder (ViT-B/32), resulting in a lower-dimensional embedding in $\mathbb{R}^{d_2}$, where $d_2 = 512$ for this clip model.

Notably, compared to the raw pixel representation, the CLIP-embedding latent space provides features that are semantically aligned with human-interpretable concepts. This is because the CLIP model [8] is trained using natural language supervision, encouraging the embedding space to capture high-level abstractions such as object categories, actions, and attributes rather than low-level texture or color information. As a result, distances computed in the CLIP space are more reflective of semantic similarity between samples, offering a meaningful basis for measuring distributional mismatch that aligns closely with task-relevant features. We conduct experiments using training datasets from COCO [9] and VisDrone [10], evaluating the distributional mismatch under both representations.

2) *Optimal Transport Computation*: We model both datasets as discrete distributions. That is, the empirical dis-

tributions are defined as:

$$\mu = \frac{1}{N} \sum_{i=1}^N \delta_{x_i}, \quad \nu = \frac{1}{M} \sum_{j=1}^M \delta_{y_j},$$

where δ_{x_i} denotes the Dirac measure distributed at x_i .

To compute the Wasserstein- p distance between μ and ν , we define a cost matrix $C \in \mathbb{R}^{N \times M}$, where each element is given by $C_{ij} = \|x_i - y_j\|^p$. The Wasserstein distance is then calculated as:

$$W_p(\mu, \nu) = \left(\min_{\gamma \in \Gamma(\mu, \nu)} \sum_{i=1}^N \sum_{j=1}^M \gamma_{ij} C_{ij} \right)^{1/p},$$

subject to the constraint that $\gamma \in \Gamma(\mu, \nu)$, the set of couplings with marginals μ and ν .

This setup enables us to evaluate distributional mismatch between training and ODD datasets across both raw and feature-level representations in a principled, mathematically grounded manner.

3) *Performance Comparison Metrics*: To evaluate the performance of object detection models, we employ the standard metric of mean Average Precision (mAP), which measures the area under the precision-recall curve across various confidence thresholds [33], [34]. Specifically, we use two variations: mAP@0.5 and mAP@0.5:0.95. The mAP@0.5 metric computes the average precision at a fixed intersection-over-union (IoU) threshold of 0.5, meaning a predicted bounding box is considered correct if it overlaps with the ground truth box by at least 50%. In contrast, mAP@0.5:0.95 is a more comprehensive metric that averages the AP scores over ten IoU thresholds, ranging from 0.5 to 0.95 in steps of 0.05. This stricter formulation penalizes both poor localization and uncertainty in detection, providing a more holistic view of model performance [9].

YOLOv8 adopts both mAP@0.5 and mAP@0.5:0.95 as evaluation metrics to capture the strengths and weaknesses

of its predictions [35]. The mAP@0.5 score is widely used and offers a baseline for comparison with legacy detectors, while the mAP@0.5:0.95 metric aligns with COCO evaluation standards and emphasizes precise localization. In practice, models may achieve high mAP@0.5 by detecting objects with loose bounding boxes, but may struggle under higher IoU thresholds. Thus, including both metrics ensures that object detectors are not only identifying objects accurately but also localizing them precisely, which is an essential requirement for safety. For example, in autonomous UAS landings, fine-grained detection quality directly impacts operational safety.

B. Diffusion Model-based Data Generation and Evaluation

In this subsection, we outline the process of generating and evaluating generated data to supplement limited real-world ODD samples. First, diffusion models are trained at multiple resolutions using drone footage to produce realistic generated images that reflect operational scenarios. Then, the quality of the generated data is assessed using the Frchet Inception Distance (FID), which measures the similarity between real and generated data distributions. A low FID score ensures that the generated data is both realistic and diverse, making it suitable for evaluating model performance under varied deployment conditions.

1) *Synthetic Data Generation*: When real-world ODD samples are limited, we leverage generative AI techniques to simulate realistic deployment conditions. Specifically, we train diffusion models [17], using frames extracted from drone-captured video footage. Due to computational and resource constraints, we train models at three resolutions: 64×64 , 128×128 and 256×256 . After training, we use the diffusion sampler to generate synthetic images, selecting only samples that realistically extend the operational scenarios present in the ODD data. The deonise (sampling) process of the images is shown in Figure 4.

2) *Quality Assessment of Generated Data*: To evaluate the quality of the generated data and ensure that it aligns with the operational design domain (ODD) scenarios, we employ the Frchet Inception Distance (FID) as a standard metric [36]. The FID score quantitatively assesses the similarity between real and generated datasets based on feature statistics extracted from a pretrained network. Specifically, FID is computed as:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right),$$

where (μ_r, Σ_r) and (μ_g, Σ_g) denote the mean and covariance matrices of real and generated feature distributions, respectively. The FID score can also be interpreted as an approximation of the 2-Wasserstein distance between two Gaussian distributions fitted to the feature embeddings of the two datasets. Lower FID scores indicate higher fidelity and diversity in the generated samples. By maintaining a low FID, we ensure that the generated data serves as a reliable extension of the ODD domain, enabling robust evaluation of the model.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

In this study, we utilize the YOLOv8 object detection model [35] to assess model performance in autonomous UAS landing scenarios. The architecture comprises three key components: a CSPDarknet-based backbone [37] for hierarchical feature extraction, a PANet-style neck [38] for multi-scale feature fusion, and an anchor-free detection head for bounding box regression and classification. During training, input images were uniformly resized to 736×736 pixels. Models were trained for 300 epochs using the Adam optimizer with cosine learning rate scheduling. To enhance generalization, we applied extensive data augmentation strategies, including mosaic blending, MixUp [39], HSV color perturbation, and random erasing [40]. Performance was evaluated using standard object detection metrics, mAP@0.5 and mAP@0.5:0.95, on real-world drone video frames annotated with vehicles and pedestrians. This configuration ensures reproducibility and provides a consistent baseline for comparison under varying training distributions.

To evaluate model performance in realistic deployment conditions, we collected a custom UAS flight test dataset using a DJI Mini 3 drone equipped with an RC-N1 controller as our ODD dataset. The flight video was recorded at 1920×1080 resolution and 30 frames per second (FPS), simulating an autonomous package delivery landing in a suburban neighborhood environment. The test scene includes a designated landing site marked with ArUco fiducial markers and an orange circular pad, surrounded by dynamic obstacles including four pedestrians and one vehicle. The video dataset spans approximately 12 videos of real-world scenes, including approximately 13,560 frames to construct the ODD dataset. These objects exhibit naturalistic motion, such as crossing into the landing zone or walking nearby, designed to replicate edge-case behaviors encountered in real-world operations.

All training and evaluation experiments are conducted on a dedicated server equipped with NVIDIA RTX 3090 GPUs, enabling high-resolution image processing and large-batch training. The system was powered by an AMD Ryzen 9 5950X 16-core processor with a base clock speed of 3.4 GHz, supported by 128 GB of DDR4 RAM. The operating system environment was Ubuntu 20.04 LTS.

B. Distributional Mismatch and Performance Comparison

We evaluate the relationship between distributional mismatch and model performance by training YOLO-based object detection models on two datasets: COCO [9] and VisDrone [10]. Each trained model is then evaluated on a fixed operational design domain (ODD) dataset captured from real deployment environments. To quantify distributional mismatch between the training and ODD datasets, we compute both 1-Wasserstein (W_1) and 2-Wasserstein (W_2) distance, by using the `ot.emd2()` function from the Python Optimal Transport (POT) library [21], which solves the discrete optimal transport problem in an efficient approach.

For this analysis, we consider two data representations: (i) raw pixel-level images and (ii) feature vectors projected into a 512-dimensional latent space using a pretrained CLIP model (ViT-B/32 backbone) [8]. For raw data, each image is treated as a high-dimensional point in pixel space (e.g., $64 \times 64 \times 3 = 12,288$ dimensions), while the CLIP projection reduces each image to a 512-dimensional semantic embedding.

To make the computation of the Wasserstein distance balanced, we sample 5,000 data from COCO dataset and VisDrone dataset, respectively, and collect all data from ODD dataset, which contains 13,560 data. In both raw pixel and CLIP embedding representations, we observe that the VisDrone dataset exhibits smaller Wasserstein distances to the ODD dataset than the COCO dataset, indicating a closer distributional mismatch. Table I summarizes these results.

TABLE I
1-WASSERSTEIN AND 2-WASSERSTEIN DISTANCES BETWEEN TRAINING DATASETS (COCO, VISDRONE) AND ODD DATA UNDER RAW AND CLIP-EMBEDDING FORMATS.

Format	Train Dataset	W_1 Distance	W_2 Distance
Raw Pixels	VisDrone	20.583	22.794
Raw Pixels	COCO	26.585	27.774
CLIP Embedding	VisDrone	0.761	0.625
CLIP Embedding	COCO	0.904	0.853

Meanwhile, we evaluate the performance of each YOLO model, trained on COCO and VisDrone datasets individually, using mean Average Precision (mAP) as the primary evaluation metric on the ODD dataset. As shown in Table II, the VisDrone-trained model achieves better performance compared to the COCO-trained model, which is consistent with our hypothesis that larger distributional mismatch correlates with lower model performance in the target domain.

TABLE II
YOLO MODEL PERFORMANCE (mAP) ON ODD DATA WHEN TRAINED ON DIFFERENT SOURCE DATASETS.

Training Dataset	mAP@0.5	mAP@0.5:0.95
COCO	50.456%	35.167%
VisDrone	95.881%	76.121%

C. Diffusion Model-based Data Generation and Evaluation

1) *Synthetic Image Generation and FID Evaluation:* To simulate ODD conditions when real samples are limited, we train diffusion models using frames extracted from drone footage. We experiment with three spatial resolutions: 64×64 , 128×128 , and 256×256 . Each diffusion model is trained for 500 epochs with 64 feature channels. To accommodate GPU memory constraints, we use a batch size of 32 for 64×64 and 128×128 resolutions, and a reduced batch size of 8 for 256×256 . Examples of generated images at different resolutions are shown in Figure 5 and Figure 6.

From each trained diffusion model, we sample 3,000 generated images and evaluate their quality using the Frchet Inception Distance (FID). Table III reports the FID scores between generated images and the real ODD images.

TABLE III
FID SCORES BETWEEN GENERATED DATA AND ODD SAMPLES. LOWER FID INDICATES HIGHER REALISM AND DIVERSITY.

Resolution	FID Score
64×64	180.532
128×128	139.729
256×256	57.294

2) *ODD Augmentation:* By augmenting the ODD dataset with synthetic images generated by the diffusion model, we mitigate the challenge of limited ODD data availability. A total of 3,000 images at a resolution of 256×256 are sampled from the trained diffusion model, with visually unrealistic samples removed through manual filtering. The distributional mismatch between the augmented ODD dataset and each training dataset (COCO and VisDrone) is then recomputed. As shown in Table IV, the augmented dataset yields more refined Wasserstein distance values compared to the original ODD set. This augmentation improves the representational diversity of the ODD domain and facilitates more robust and reliable evaluation of model performance.

TABLE IV
1-WASSERSTEIN AND 2-WASSERSTEIN DISTANCE BETWEEN TRAINING DATASETS (COCO, VISDRONE) AND ODD DATA WITH GENERATED DATA UNDER RAW AND CLIP-EMBEDDING FORMATS.

Format	Train Dataset	W_1 Distance	W_2 Distance
Raw Pixels	VisDrone	18.458	20.643
Raw Pixels	COCO	23.332	25.654
CLIP Embedding	VisDrone	0.641	0.532
CLIP Embedding	COCO	0.847	0.813

V. DISCUSSION

Our findings highlight a promising direction for estimating model performance under distributional mismatch using tools from optimal transport and generated data generation. The observed correlation between Wasserstein distances and performance degradation offers a principled, data-driven means of anticipating model performance in safety-critical applications such as autonomous landings. Moreover, by leveraging generative models, particularly diffusion models, we demonstrate the potential to supplement limited ODD data with realistic generated samples to extend the operational evaluation space.

Despite the progress, several limitations persist. In practice, collecting high-quality ODD data is inherently challenging due to the high cost involved, resulting in a scarcity that hinders thorough evaluation of real-world generalization. Additionally, the generated scenarios are uncontrollable. Future work could investigate adaptive generative approaches informed by domain knowledge or leverage active sampling strategies to more effectively capture representative ODD data.

VI. CONCLUSION

In this work, we proposed a framework for comparing model performance under distributional mismatch by leveraging optimal transport theory. By quantifying the mismatch

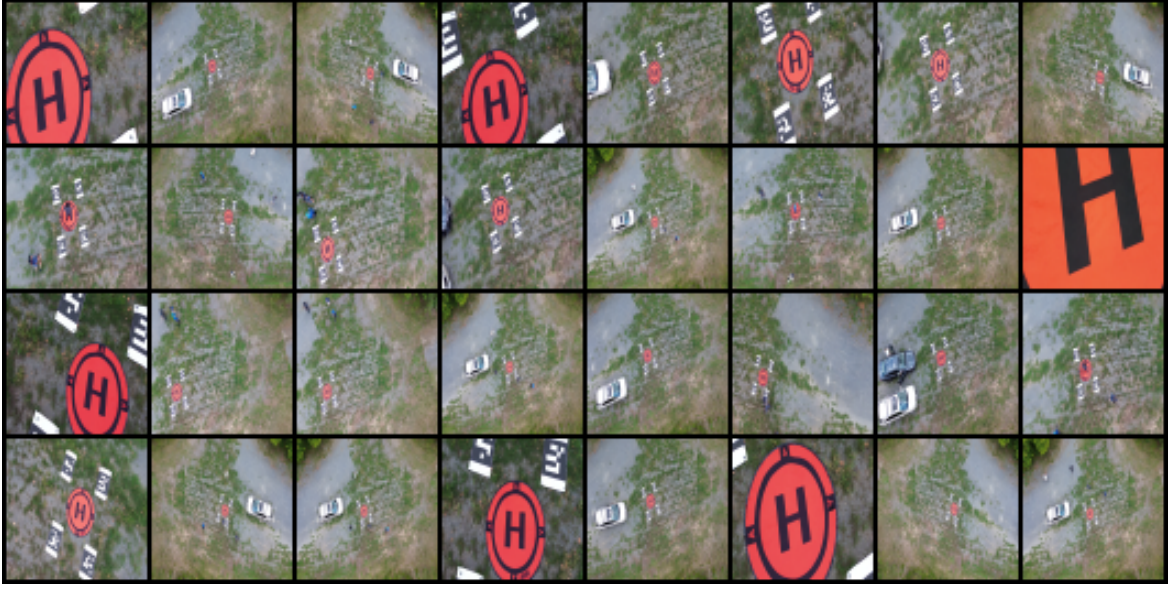


Fig. 5. This figure presents synthetic images produced by our diffusion model at a resolution of 64×64 . The generated samples resemble aerial views featuring helipad markings and surrounding terrain, closely aligning with the visual characteristics of the ODD dataset. This indicates that the model has successfully captured the underlying data distribution.



Fig. 6. The left side of the figure displays the generated images at a resolution of 256×256 , while the right side shows the same scenes with object detection results overlaid. The successful detection of objects in the synthetic images by our YOLO model, trained on COCO and fine-tuned on VisDrone, demonstrates both the reality and utility of the generated data.

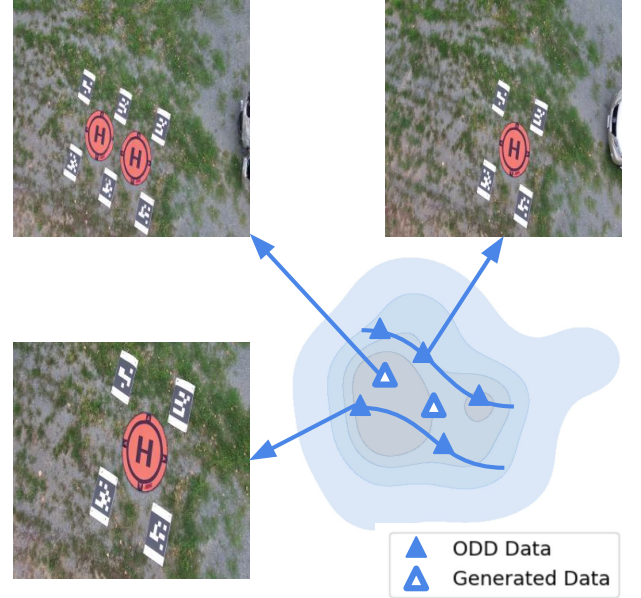


Fig. 7. Illustration of the interpolation effect of synthetic data in the ODD domain. Solid and hollow blue triangles represent real and synthetic data points, respectively, linked to their corresponding images. The background contour map reflects a conceptual high-dimensional embedding space, where blue trajectories denote continuous frame sequences from drone-captured aerial videos. Synthetic samples interpolate between these trajectories, bridging gaps across data streams. This "bootstrap effect" enriches the ODD distribution, enabling broader coverage and more robust model evaluation and training.

between training datasets and operational design domain (ODD) samples using Wasserstein distances, we demonstrated that distributional mismatch strongly correlates with model degradation in the target domain. To address the scarcity of

real-world ODD data, we trained diffusion models to generate images and validated their quality using Frchet Inception Distance (FID). Our experiments on object detection tasks

showed that models trained on datasets closer in distribution to the ODD domain exhibit superior performance, validating our hypothesis.

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